

Topic 18: Organic Chemistry III

In order to develop their practical skills, students should be encouraged to carry out a range of practical experiments related to this topic. Possible experiments include investigating the reactions of different functional groups, preparing an aromatic ester such as methyl benzoate, making nylon, purifying an organic solid.

Mathematical skills that could be developed in this topic include calculating the resonance stability of benzene from thermodynamic data, calculating percentage yields.

Within this topic, students can consider how the model for benzene structure has developed in response to new evidence. By this stage, their continuing practical experience should enable them to use techniques to carry out reactions and purify products efficiently and safely.

Students should:

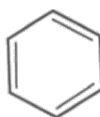
Topic 18A: Arenes - benzene

1. understand that the bonding in benzene has been represented using the Kekulé and the delocalised model, the latter in terms of overlap of p -orbitals to form π -bonds
2. understand that evidence for the delocalised model of the bonding in benzene is provided by data from enthalpy changes of hydrogenation and carbon-carbon bond lengths

Students may represent the structure of benzene as:



or



as appropriate in equations and mechanisms.

3. understand why benzene is resistant to bromination, compared with alkenes, in terms of delocalisation of π -bonds in benzene and the localised electron density of the π -bond in alkenes
4. understand the reactions of benzene with:
 - i oxygen in air (combustion with a smoky flame)
 - ii bromine, in the presence of a catalyst
 - iii a mixture of concentrated nitric and sulfuric acids
 - iv halogenoalkanes and acyl chlorides with aluminium chloride as catalyst (Friedel-Crafts reaction)
5. understand the mechanism of the electrophilic substitution reactions of benzene (halogenation, nitration and Friedel-Crafts reactions), including the generation of the electrophile
6. understand the reaction of phenol with bromine water